

### Colligative properties of electrolyte

- (c)  $(NH_4)_3PO_4$  gives maximum ion. Hence, its osmotic pressure is maximum.
- (b)  $BaCl_2$  gives maximum ion hence it shows lowest vapour pressure.
- (d)  $Na_3PO_4$  consists of maximum ions hence it shows lowest vapour pressure.  
 $Na_3PO_4 \rightarrow 3Na^+ + PO_4^{3-} = 4 \text{ ion.}$
- (c) Vapour pressure of a solvent is lowered by the presence of solute in it. Lowering in vapour pressure is a colligative property *i.e.*, it depends on the no. of particles present in the solution.  $Cu(NO_3)_2$  gives the maximum no. of ions. (*i.e.*, 3) so it causes the greatest lowering in vapour pressure of water.
- (c)  $Na_2SO_4$  has more osmotic pressure than  $NaCl$  solution because  $Na_2SO_4$  gives 3 ions.
- (a)  $NaCl$  gives maximum ion hence it will show highest osmotic pressure.
- (c)  $M/10 BaCl_2$

**Explanation:**

pressure:

$$\pi = iCRT$$

Higher the van't Hoff factor  $i$ , higher the osmotic pressure.

- Urea  $\rightarrow$  non-electrolyte  $\rightarrow i = 1$
- Glucose  $\rightarrow$  non-electrolyte  $\rightarrow i = 1$
- $HCl \rightarrow$  dissociates into 2 ions  $\rightarrow i = 2$
- $BaCl_2 \rightarrow Ba^{2+} + 2Cl^- \rightarrow$  total 3 ions  $\rightarrow i = 3$

Therefore  $BaCl_2$  has maximum osmotic pressure.

- (c)  $BaCl_2 \Rightarrow Ba^{2+} + 2Cl^- = 3 \text{ ion}$   
 $NaCl \Rightarrow Na^+ + Cl^- = 2 \text{ ion}$   
 Glucose  $\Rightarrow$  No ionisation



$\therefore BaCl_2 > NaCl > \text{Glucose}$

9. (a)  $Al_2(SO_4)_3$  gives maximum osmotic pressure because it gives 5 ion.
10. (a) Highest osmotic pressure is given by solution which produce maximum number of ions i.e.  $CaCl_2$ .
11. (c)  $BaCl_2$  gives maximum ion. Hence, it shows highest boiling point.
12. (c)  $BaCl_2$  gives maximum ion. Hence, its boiling point is maximum.
13. (d)  $CaCl_2$  gives maximum ion hence it shows highest boiling point.
14. (c) Elevation in boiling point is a colligative property which depends upon the number of solute particles. Greater the number of solute particle in a solution higher the extent of elevation in boiling point.  
 $Na_2SO_4 \rightarrow 2Na^+ + SO_4^{2-}$
15. (a)  $NaCl$  contain highest boiling point than other's compound.
16. (d)  $KCl > C_6H_5OH > C_6H_{12}O_6$   
 Boiling point decreasing order  $\rightarrow$   
 chloride is ionic compound and phenol is formed phenoxide ion hence it shows greater boiling point than glucose.
17. (c)  $Al_2(SO_4)_3$  gives maximum ion hence it will show highest boiling point.
18. (b)  $NaCl$  is a more ionic compare to  $BaCl_2$ , glucose and urea solution.
19. (c) Urea = 1 ; Common salt = 1 ;  $Na_2SO_4$  = 3  
 Ratio = 1 : 2 : 3
20. (c)  $CaCl_2$  gives maximum ion hence it has minimum freezing point.

